



MATEMATICI 2

Curs 12

Exemplu de determinare a complementului ortognal

$V = \mathbb{R}^3$, $W = L(v_1 = (1, 2, 3))$

$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle =$

$= x_1 y_1 + x_2 y_2 + x_3 y_3$

$W^{\perp} = ?$

Rez.: Fie $B_W = \{v_1 = (1,2,3)\}$

$B_V = \{ v_1 = (1,2,3) , v_2 = (1,0,0) , v_3 = (0,1,0) \}$

Liniar independența vectorilor $\Leftrightarrow$

$\Leftrightarrow$ liniar independența coord. la o bază $\Leftrightarrow$

$\Leftrightarrow$ liniar indep. liniilor unei matrici care conț. coord.

Minorul principal din matricea coord. conț. coord. unor vectori dintr-o submulțime maximală liniar indep.

Dc. rangul matricei = nr. vectorilor atunci vectorii sunt liniar indep.

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 3 & 0 & 0 \end{pmatrix}$$

$\det(A) = 3 \Rightarrow \{v_1, v_2, v_3\}$ liniar independenți.

+ bară pt. V

Ortogonalizăm baza B_V cu Gram-Schmidt.

$u_1 = v_1 = \{1, 2, 3\}$

$u_2 = v_2 + \alpha u_1$

$\langle u_2, u_1 \rangle = 0$

$\langle v_2 + \alpha u_1, u_1 \rangle = \langle v_2, u_1 \rangle + \alpha \langle u_1, u_1 \rangle = 0 \Rightarrow$

$$\Rightarrow t = \frac{-\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} = \frac{-\langle (1,0,0), (1,2,3) \rangle}{\langle (1,2,3), (1,2,3) \rangle} = \frac{-1}{14}$$

$$u_2 = v_2 - \frac{u_1}{14} = \langle 1,0,0 \rangle - \frac{1}{14} \langle 1,2,3 \rangle =$$

$$= \left\langle \frac{13}{14}, \frac{-1}{7}, \frac{-3}{14} \right\rangle$$

$u_3 = v_3 + \beta u_1 + \gamma u_2$

$\langle u_3, u_1 \rangle = 0$

$\langle v_3 + \beta u_1 + \gamma u_2, u_1 \rangle = \langle v_3, u_1 \rangle + \beta \langle u_1, u_1 \rangle +$

$+ \gamma \underbrace{\langle u_2, u_1 \rangle}_{0} = 0 \Rightarrow$

$$\Rightarrow \beta = \frac{-\langle v_3, u_1\rangle}{\langle u_1, u_1\rangle} = \frac{-\langle (0,1,0), (1,2,3)\rangle}{\langle (1,2,3), (1,2,3)\rangle} =$$

$$= \frac{-2}{14} = \frac{-1}{7}$$

$$\langle u_3, u_2\rangle = 0$$

$$\langle v_3 + \beta u_1 + \gamma u_2, u_2\rangle = \langle v_3, u_2\rangle + \beta \underbrace{\langle u_1, u_2\rangle}_{0} +$$

$$+ \gamma^2 \langle u_2, u_2\rangle = 0 \Rightarrow$$

$$\Rightarrow \gamma = \frac{-\langle v_3, u_2\rangle}{\langle u_2, u_2\rangle}$$

$$\gamma = \frac{-\langle (0,1,0), \left(\frac{13}{14}, \frac{-2}{14}, \frac{-3}{14}\right)\rangle}{\langle \left(\frac{13}{14}, \frac{-2}{14}, \frac{-3}{14}\right), \left(\frac{13}{14}, \frac{-2}{14}, \frac{-3}{14}\right)\rangle}$$

$$\gamma^2 = \frac{\frac{2}{14}}{\frac{13\cdot 13}{14\cdot 14} + \frac{2\cdot 2}{14\cdot 14} + \frac{3\cdot 3}{14\cdot 14}} = \frac{2}{14\left(\frac{169+4+9}{14\cdot 14}\right)} =$$

$$= \frac{2\cdot 14}{182} = \frac{14}{91}$$

$$\gamma^2 = \frac{14}{91}$$

$$u_3 = v_3 - \frac{\langle 1,2,3\rangle}{14}\langle 1,2,3\rangle + \frac{28}{182}\left(\frac{13}{14}, \frac{-2}{14}, \frac{-3}{14}\right) = \dots$$

$$W^{\perp} = L(u_2, u_3)$$

Temă: Fie V sp. vect. finit înzestrat cu prod. scalar;

$W, U \subseteq V$.

Să se arate că $W \perp U \iff$

$\iff w_i \perp u_j \quad i = \overline{1,q}$, $j = \overline{1,p}$

unde $\{w_1, \dots, w_q\} = B_W$ bază a lui W

$\{u_1, \dots, u_p\} = B_U$ bază a lui U

Baza duală a unui spațiu vectorial

Fie V = sp. vect. finit peste K cu baza

$B_V = \{e_1, \ldots, e_n\}$

Cons. că corpul K, având ca lege int. + scalarilor și ca lege ext. înm. a doi scalari, formează o structură de sp. vectorial pt. peste K.

$\dim_K K = 1$ pt. că $B_K = \{1\}$

↓

elem. neutru în rap. cu înm.

funcțională = $\forall$ apl. liniară $f: V \to K$

$\mathrm{Hom}(V, K) = \{ f : V \to K \mid f = \text{liniară} \}$

Dc. $B_V = \{e_1, \dots, e_n\}$

$B_K = \{1\}$

$M_f^{B_V B_K}$ are $\in M_{n,1}(K)$

Apl. $F : Hom(V, K) \to M_{n,1}(K)$

$F(f) = M_f^{B_V B_K}$

este izomorfism de sp. vectoriale considerând

$M_{n,1}(K)$ = sp. vect. peste K cu

legea int. ad. matricilor și legea ext. înm. unui

scalar cu o matrice.

$\Rightarrow \dim_K \mathrm{Hom}(V, K) = \dim_K M_{n,1}(K) = n$

Def. o bază în $\mathrm{Hom}(V, K)$.

Baza duală a unei baze din V.

Fie $B_V = \{e_1, \dots, e_n\}$.

Cons. în $\mathrm{Hom}(V, K)$ mult. de vectori $B^* = \{f_1, f_2, \dots, f_n\}$

cu prop. $f_i(e_j) = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$

liniar indep.

Fie $\alpha_1 f_1 + \dots + \alpha_n f_n = 0$, $0 =$ functionala nulă

$(\alpha_1 f_1 + \dots + \alpha_n f_n)(e_i) = 0(e_i)$

$\alpha_1 \underbrace{f_1(e_i)}_{0} + \dots + \alpha_i \underbrace{f_i(e_i)}_{1} + \dots + \alpha_n \underbrace{f_n(e_n)}_{0} = 0 \Rightarrow$

$\Rightarrow \alpha_i = 0, \quad i = \overline{1,n}$

$\Rightarrow B^*$ = liniar indep. $\Rightarrow B^*$ = bază în $Hom(V, K)$

Teoremă (Riesz)

Fie V = sp. vectorial, finit, real sau complex ($K = \mathbb{R}$ sau $\mathbb{C}$)

Dc. $f \in \text{Hom}(V, K) \Rightarrow$

$\Rightarrow \exists! v_0 \in V$ a.î. $f(v) = \langle v, v_0 \rangle$

Adjunctul unei aplicații liniare

Fie V = sp. vect. finit înzestrat cu prod. scalar peste $\mathbb{R}$ sau $\mathbb{C}$

Dc. $f : V \to V$ apl. liniară (adică f endomorfism al lui V)

Not. $f \in End(V)$

atunci construim funcționala f

$$F_y(x) = \langle f(x), y \rangle$$

$$F_y : V \to K$$

y = fixat

(Y). Th. Riesz $\Rightarrow$ $\exists!\ v_0 \in V$ a.î. $F_y(x) = \langle x, v_0 \rangle$

Cum v_0 depinde de y și de f, notăm

$v_0 = f^*(y) \Rightarrow$

$\Rightarrow F_y(x) = \langle f(x), y \rangle = \langle x, f^*(y) \rangle$

$= \langle x, v_0 \rangle$

Vom arăta că f^* e apl. liniară.

$\langle x, f^*(\alpha y_1 + \beta y_2)\rangle = \langle f(x), \alpha y_1 + \beta y_2\rangle =$

$= \overline{\alpha}\langle f(x), y_1\rangle + \overline{\beta}\langle f(x), y_2\rangle =$

$= \overline{\alpha}\langle x, f^*(y_1)\rangle + \overline{\beta}\langle x, f^*(y_2)\rangle$

$= \langle x, \alpha f^*(y_1)\rangle + \langle x, \beta f^*(y_2)\rangle =$

$= \langle x, \alpha f^*(y_1) + \beta f^*(y_2)\rangle, \forall x \in V \Rightarrow$

$\Rightarrow f^*(\alpha y_1 + \beta y_2) = \alpha f^*(y_1) + \beta f^*(y_2) \Rightarrow$

$\Rightarrow f^*$ apl. liniară

f^* sn. ADJUNCTA lui f

Temă:

Dc. $\langle v, u_1 \rangle = \langle v, u_2 \rangle$, $\forall v \in V \Rightarrow u_1 = u_2$

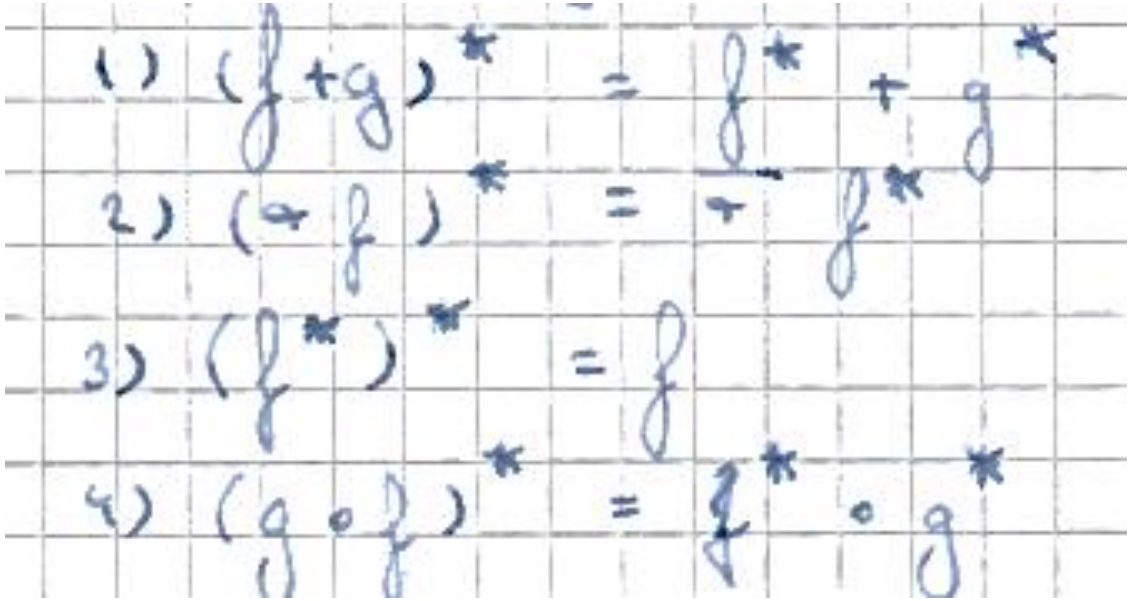
$\langle v, u_1 \rangle - \langle v, u_2 \rangle = 0$

$+ \langle u_1, v \rangle - \langle u_2, v \rangle = 0$

$\langle u_1 - u_2, v \rangle = 0$, $\forall v \in V \Rightarrow \langle v, u_1 - u_2 \rangle = 0, \forall v \in V$

$\Rightarrow u_1 - u_2 = 0 \Rightarrow u_1 = u_2$

Proprietăți ale adjunctei



5) $(1_V)^* = 1_V$ $\quad 1_V : V \to V$; $1_V(v) = v$

6) $(0_V)^* = 0_V$

Dem:

1) $\langle (f+g)(x), y \rangle = \langle x, (f+g)^*(y) \rangle =$

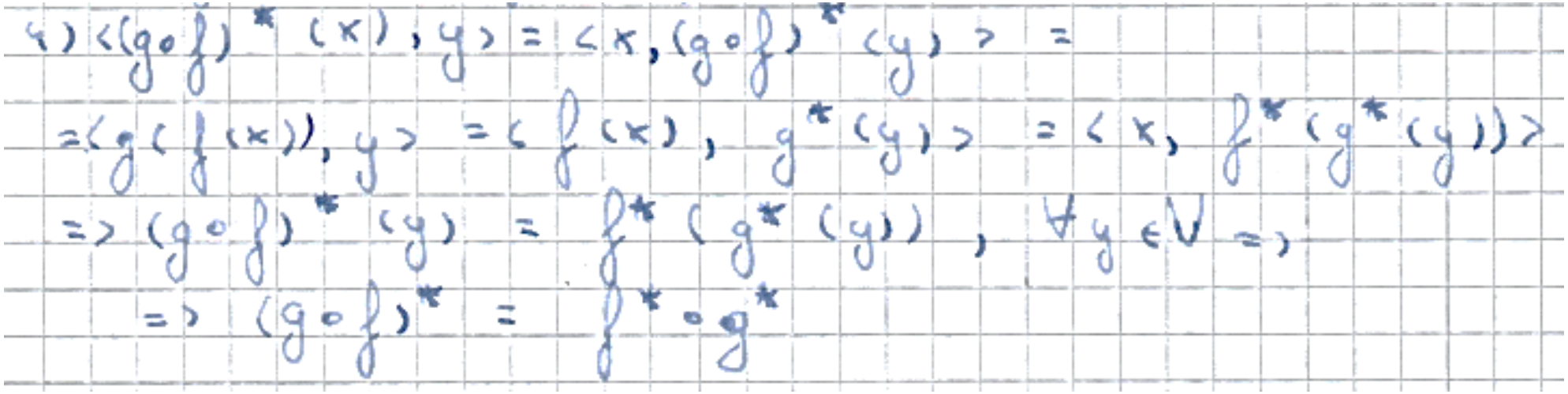
$= \langle f(x) + g(x), y \rangle = \langle f(x), y \rangle + \langle g(x), y \rangle$

$= \langle x, f^*(y) \rangle + \langle x, g^*(y) \rangle =$

$= \langle x, f^*(y) + g^*(y) \rangle = \langle x, (f^* + g^*)(y) \rangle, \forall x \in V \Rightarrow$

$\Rightarrow (f+g)^*(y) = (f^* + g^*)(y), \forall y \in V \Rightarrow$

$\Rightarrow (f+g)^* = f^* + g^*$


4) ⟨(g∘f)* (x), y⟩ = ⟨x, (g∘f)* (y)⟩ =
= ⟨g(f(x)), y⟩ = ⟨f(x), g*(y)⟩ = ⟨x, f*(g*(y))⟩
⇒ (g∘f)* (y) = f*(g*(y)), ∀ y ∈ V ⇒
⇒ (g∘f)* = f* ∘ g*